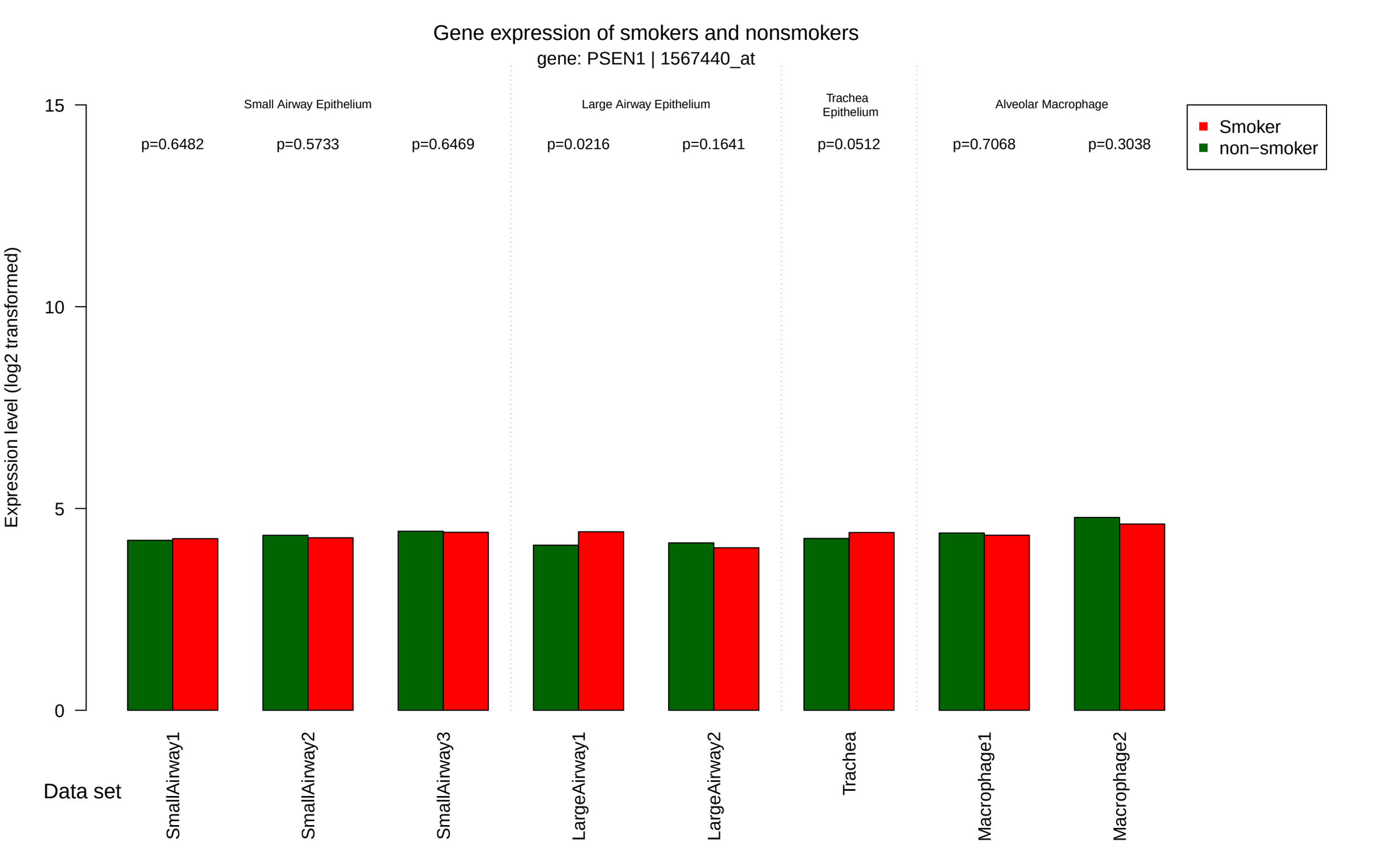
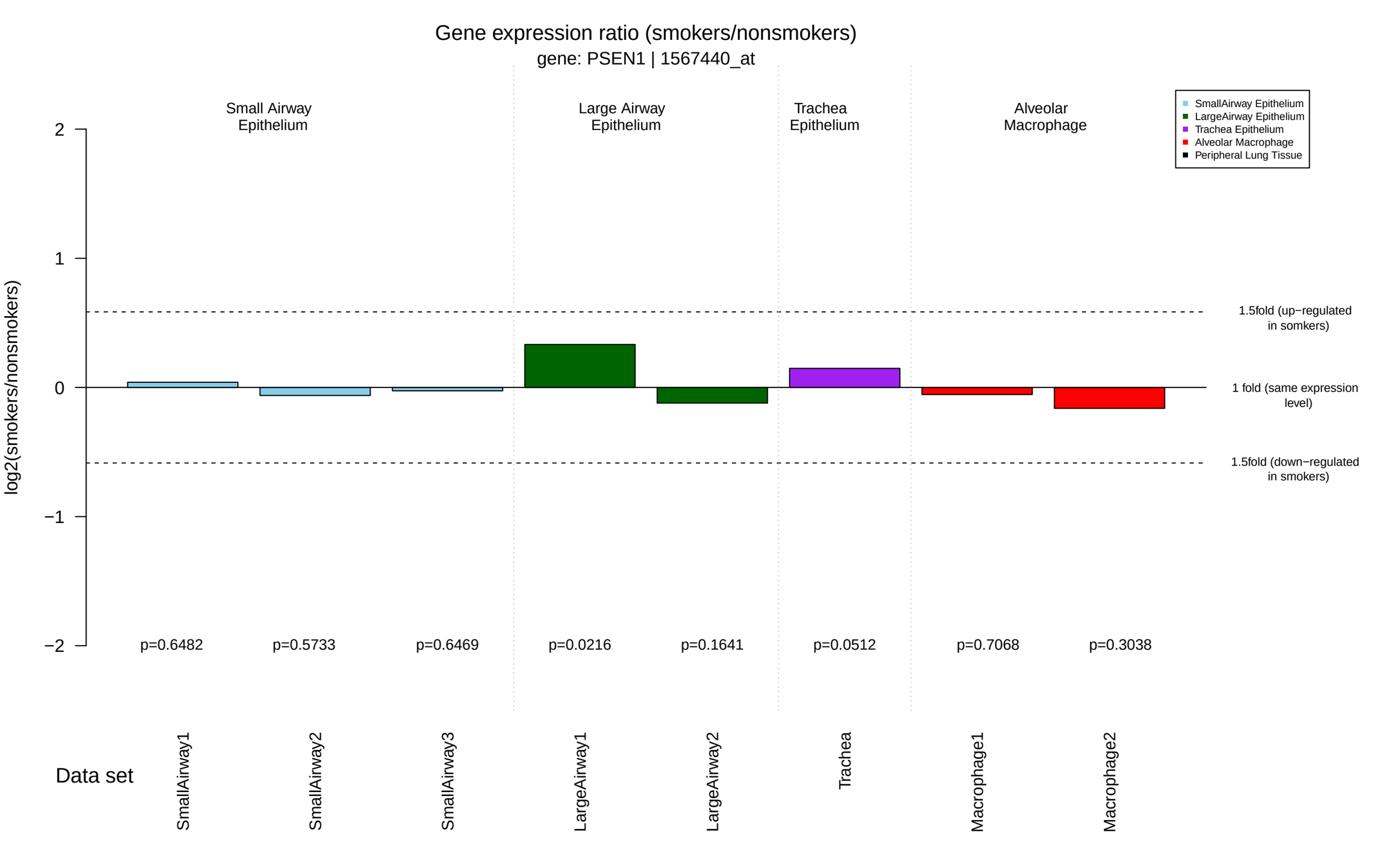
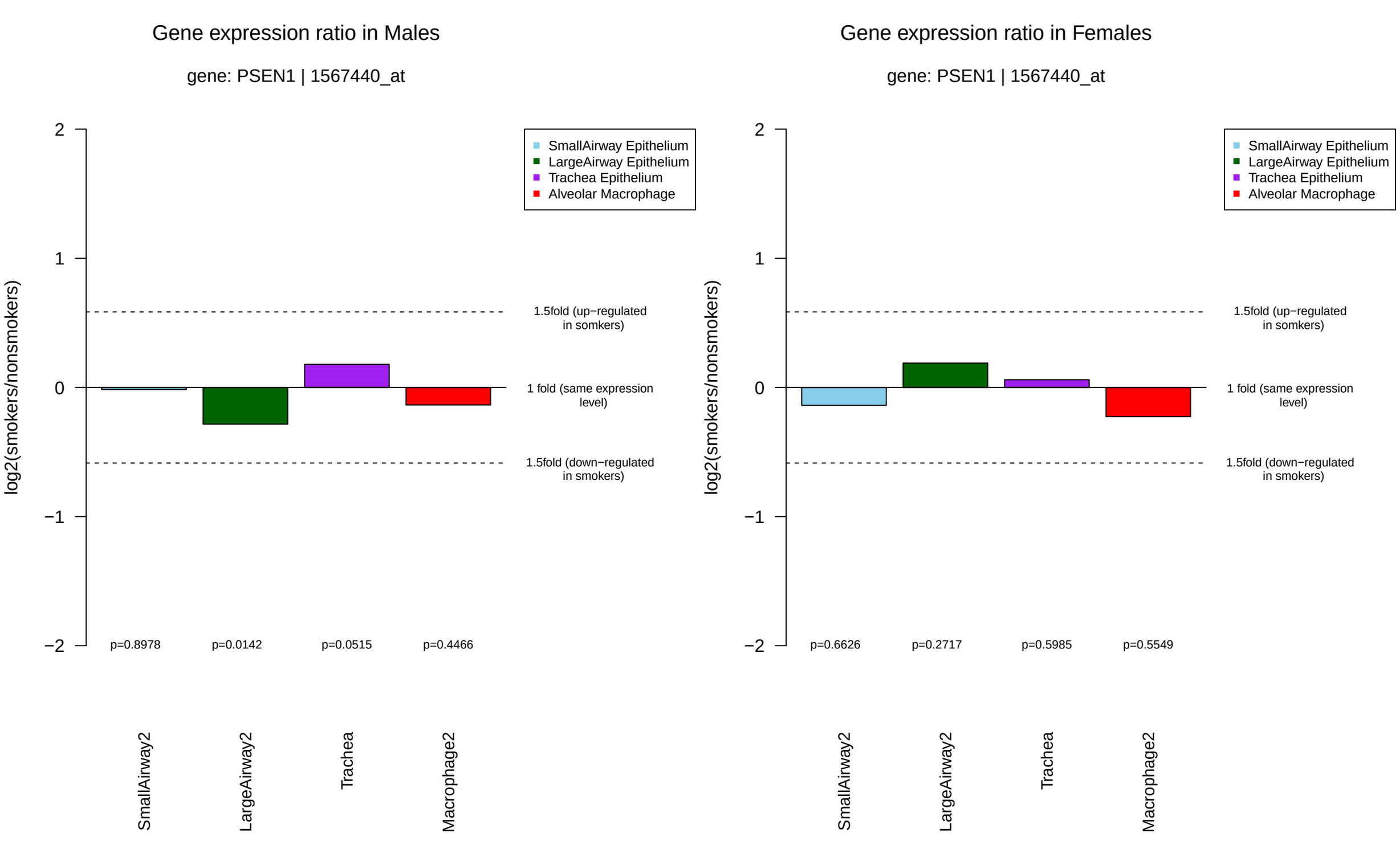
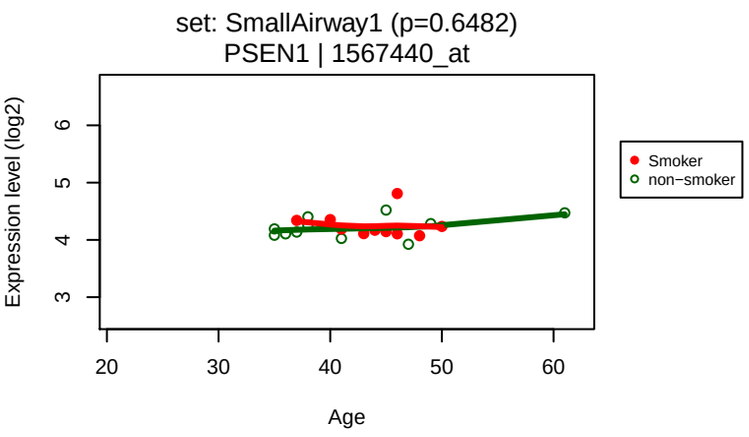
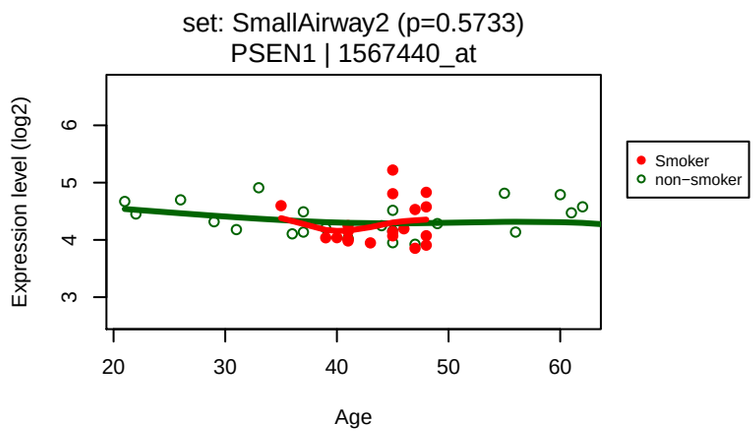
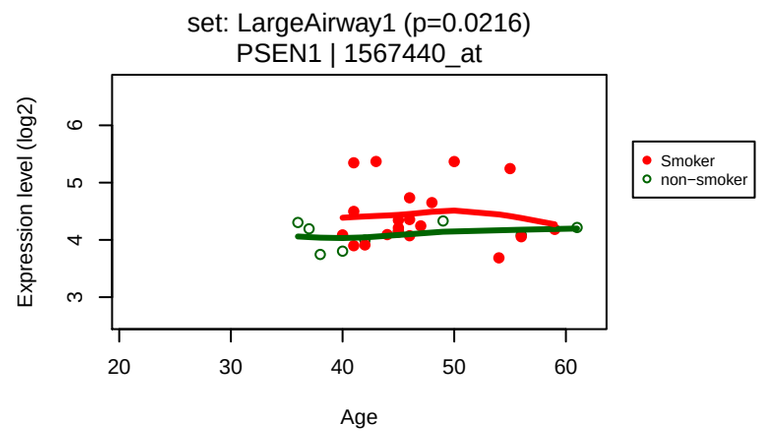

Gene expression of smokers and nonsmokers
gene: PSEN1 | 1567440_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6482
p=0.5733
p=0.6469
p=0.0216
p=0.1641
p=0.0512
p=0.7068
p=0.3038
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

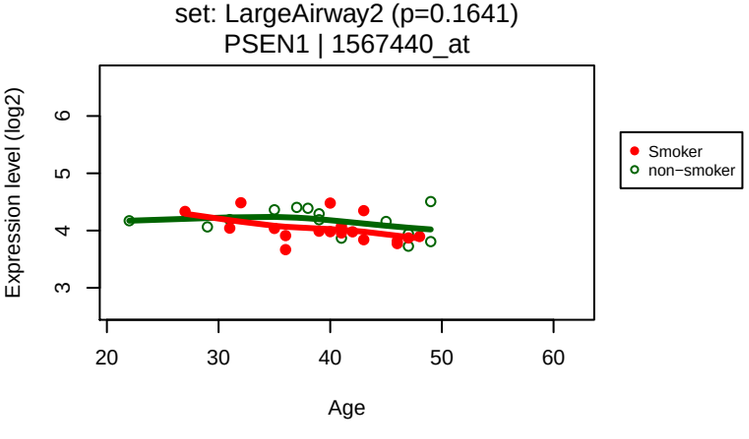

Gene expression ratio (smokers/nonsmokers)
gene: PSEN1 | 1567440_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6482
p=0.5733
p=0.6469
p=0.0216
p=0.1641
p=0.0512
p=0.7068
p=0.3038
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

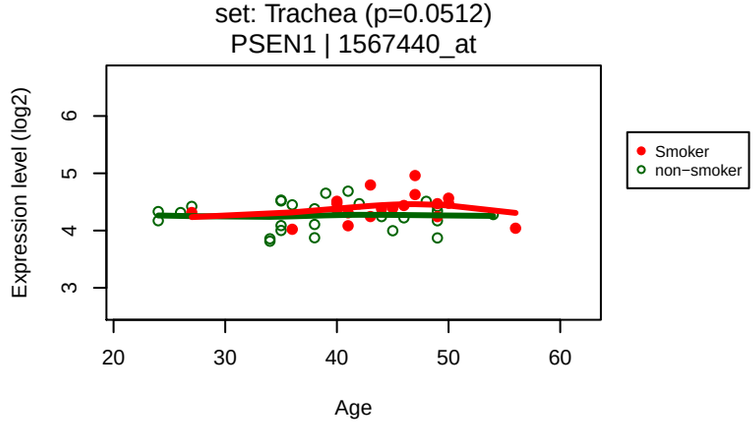

Gene expression ratio in Males
gene: PSEN1 | 1567440_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8978
p=0.0142
p=0.0515
p=0.4466
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSEN1 | 1567440_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6626
p=0.2717
p=0.5985
p=0.5549
SmallAirway2
LargeAirway2
Trachea
Macrophage2

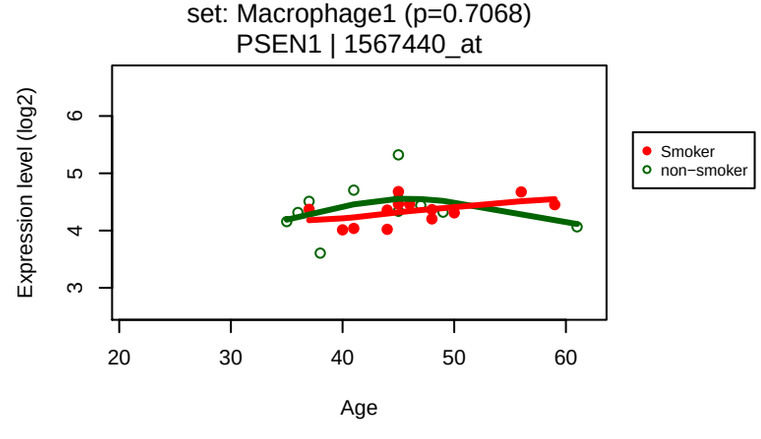

set: SmallAirway1 (p=0.6482)
PSEN1 | 1567440_at
Expression level (log2)
Age
Smoker
non−smoker

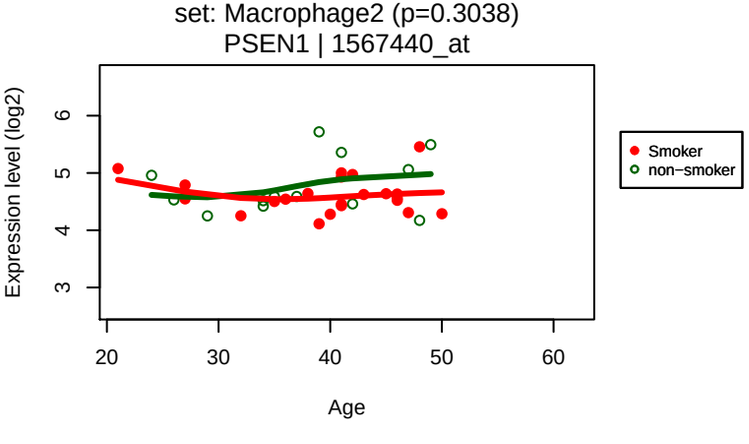

set: SmallAirway2 (p=0.5733)
PSEN1 | 1567440_at
Expression level (log2)
Age
Smoker
non−smoker

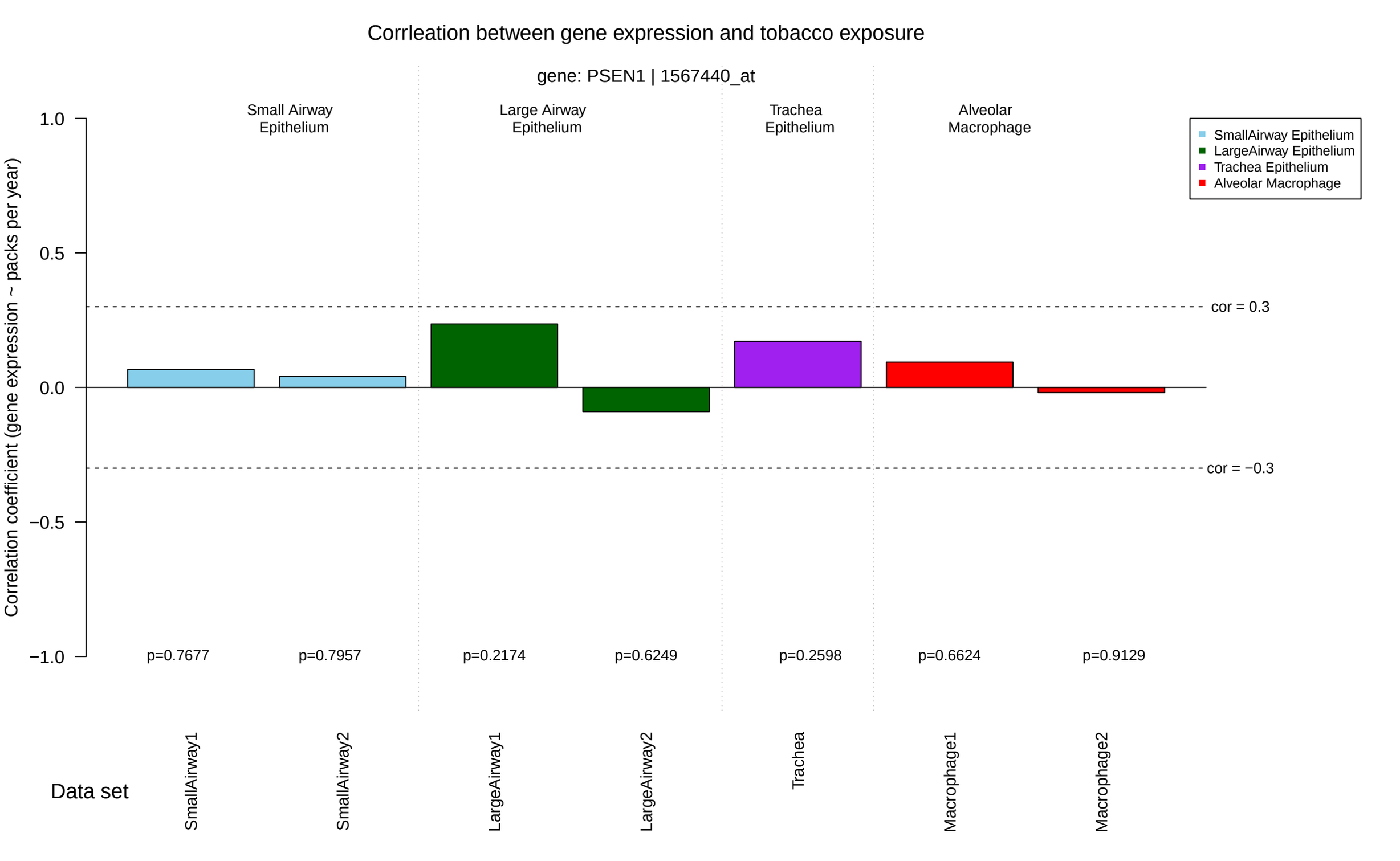

set: LargeAirway1 (p=0.0216)
PSEN1 | 1567440_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1641)
PSEN1 | 1567440_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0512)
PSEN1 | 1567440_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7068)
PSEN1 | 1567440_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3038)
PSEN1 | 1567440_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSEN1 | 1567440_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7677
p=0.7957
p=0.2174
p=0.6249
p=0.2598
p=0.6624
p=0.9129
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2